



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2017
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

15 September 2017

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
1	/ 12
2	/ 9
3	/ 9
4	/ 14
5	/ 10
6	/ 14
7	/ 12
8	/ 9
9	/ 11
Total	/ 100

This paper consists of **23** printed pages.

[Turn over]

QUESTION 1

- (a) In 1934, two biologists Davson and Danielli published their suggestion for the structure of the cell surface membrane, as shown in Fig. 1.1.

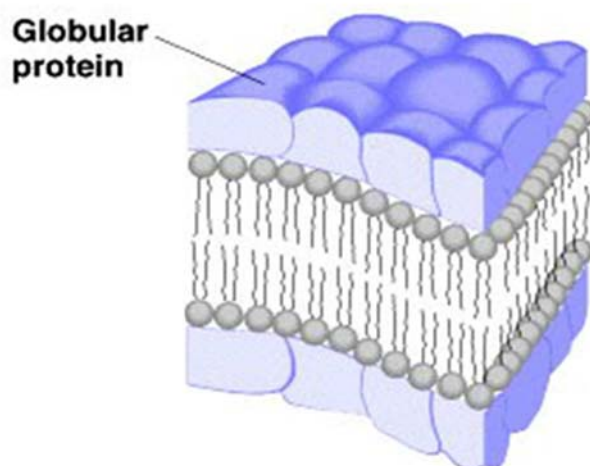


Fig. 1.1

- (i) State one way in which the Davson-Danielli structure is similar to the fluid mosaic structure and one way in which it differs from the fluid mosaic model. [2]

Similarity

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Difference

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- (ii) Suggest two problems that the Davson-Danielli structure of the membrane would pose to the functioning of the cell. [2]

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(b) Transport of substances across membranes involve many different mechanisms.

Fig. 1.2 is a diagram showing the transport of protein-rich solid particles into an animal cell.

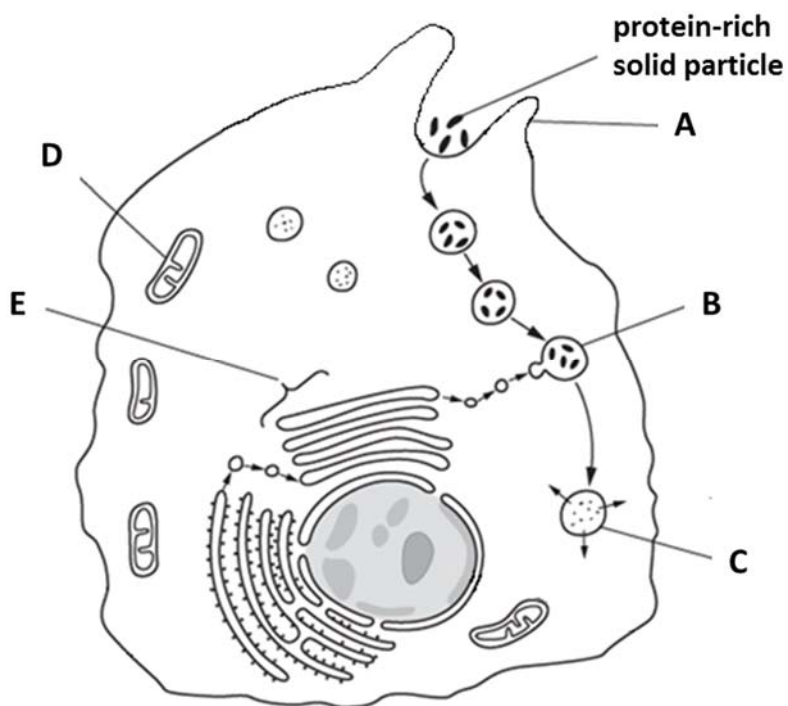


Fig. 1.2

(i) Describe the process at **A**. [2]

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(ii) Describe what happens to the protein-rich solid particle between **B** and **C**. [2]

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(iii) Name the organelles **D** and **E** and briefly describe their roles in the formation of **C**. [4]

Name of organelle D

Role of organelle D

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Name of organelle E

Roles of organelle E

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[Total: 12]

QUESTION 2

Glucose is phosphorylated at the start of glycolysis by the tetrameric enzyme, hexokinase.

There are multiple hexokinase isozymes (I-IV) for the phosphorylation of glucose, enabling specific organs to regulate carbohydrate metabolism in a unique way. Hexokinase IV, also called glucokinase, is the predominant isozyme in the liver, while hexokinase I is found in almost all other tissues.

Fig. 2.1 shows the difference in fractional saturation between glucokinase and hexokinase I, which represents the fraction of binding sites that are occupied by glucose.

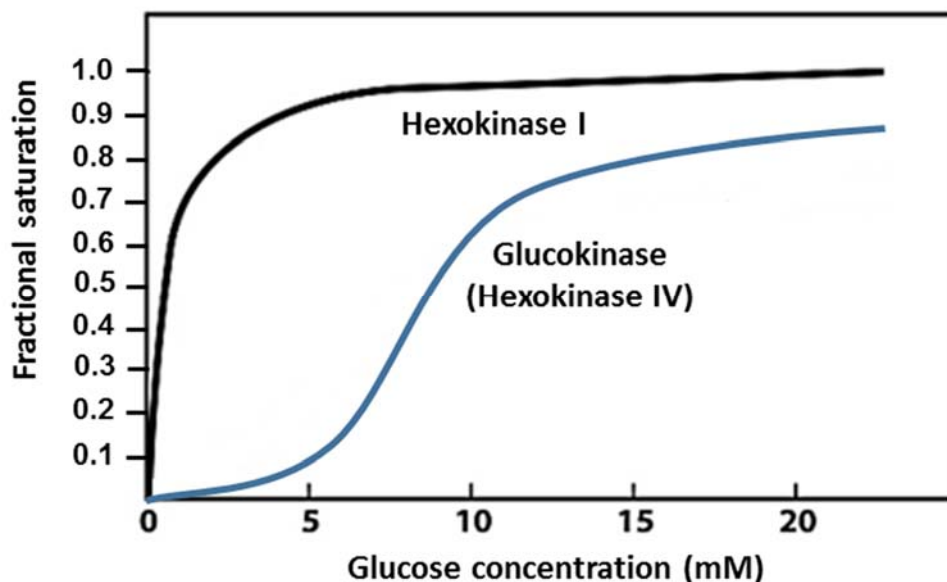


Fig. 2.1

- (a) With reference to Fig. 2.1 and using your knowledge of enzymes, account for the shape of the curve for glucokinase from 0 to 10mM of glucose concentration. [4]

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- (b)** Suggest an advantage to most cells in the body of containing hexokinase I rather than glucokinase. [2]

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- (c)** During a sporting event, muscle cells of an athlete may have to carry out respiration in anaerobic as well as aerobic conditions to produce sufficient ATP.

- (i)** Name the membrane-bound enzyme responsible for producing ATP from ADP and inorganic phosphate. [1]

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- (ii)** Explain how anaerobic respiration helps to meet the demand for sufficient ATP. [2]

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[Total: 9]

QUESTION 3

In vertebrates, sister chromatid cohesion is dependent on a complex of proteins called cohesin, which binds to and joins sister chromatids at the centromere until the onset of anaphase.

Sister chromatid separation is initiated by cleavage of cohesin by the enzyme separase. Prior to anaphase, a protein called securin binds to separase and maintains it in the inactive form. Anaphase is initiated when securin is degraded, freeing the enzyme separase.

Fig. 3.1 illustrates the transition from metaphase to anaphase during a mitotic cell cycle.

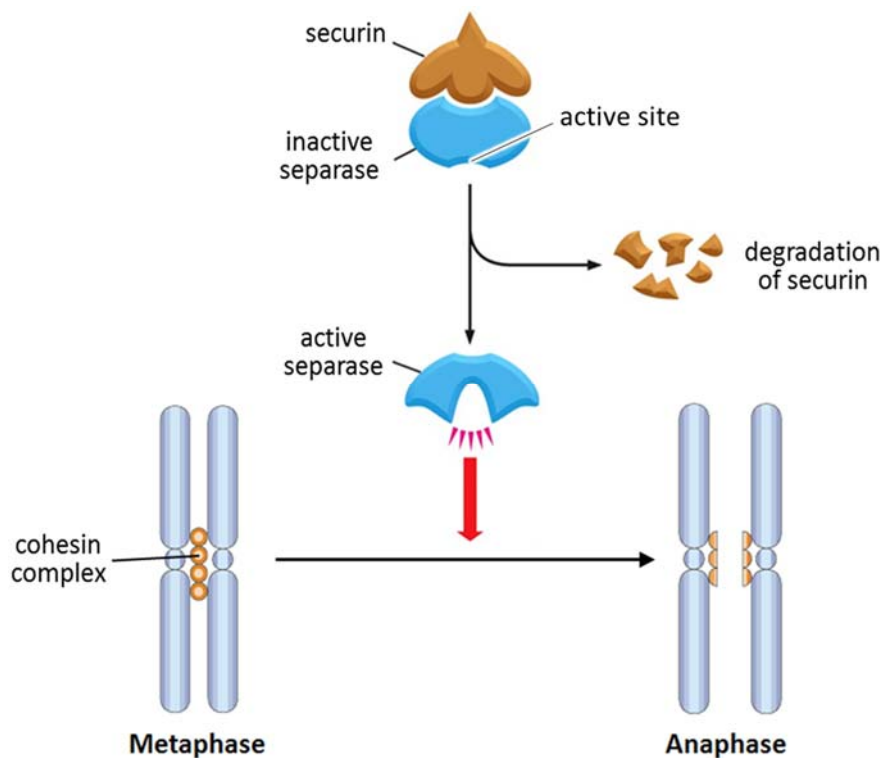


Fig. 3.1

(a) State a feature of centromeric DNA. [1]

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(b) Explain how a mutation to the centromeric DNA can lead to aneuploidy. [3]

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(c) State the class of enzyme to which separase belongs. [1]

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(d) Explain how securin maintains separase in the inactive form. [3]

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(e) Explain how securin is degraded at the onset of anaphase. [1]

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[Total: 9]

QUESTION 4

Takahashi and fellow scientists had successfully reprogrammed human fibroblasts into a pluripotent state, known as induced pluripotent stem cells (iPS cells).

- (a) Define the term *pluripotent stem cells*. [1]

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- (b) The generation of iPS cells made use of four protein factors (non-enzymatic proteins), which are introduced into differentiated cells by retroviruses. Research has proven their function in upregulating “stemness” genes, while suppressing differentiation-associated genes in human iPS cells.

- (i) State the general name given to proteins such as those four protein factors. [1]

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- (ii) Explain why the protein factor involved in upregulating “stemness” genes will contain both a DNA-binding domain and a protein-binding domain. [2]

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- (iii) Explain how an amino acid substitution within the DNA-binding domain can affect the function of the protein factor in (b)(ii). [3]

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- (iv) Explain why it is important that the scientists ensure high telomerase activities in the iPS cells. [3]

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(c) Further studies had shown that some iPS cells developed tumors, which is often attributed to the use of retrovirus. This issue of tumourgenesis must be overcome before iPS cells can be used in human therapies.

(i) Suggest why the development of tumor in the iPS cells may be attributed to the use of retrovirus. [2]

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(ii) With reference to the benefits and problems of iPS cells, discuss whether research on iPS cells should be continued. [2]

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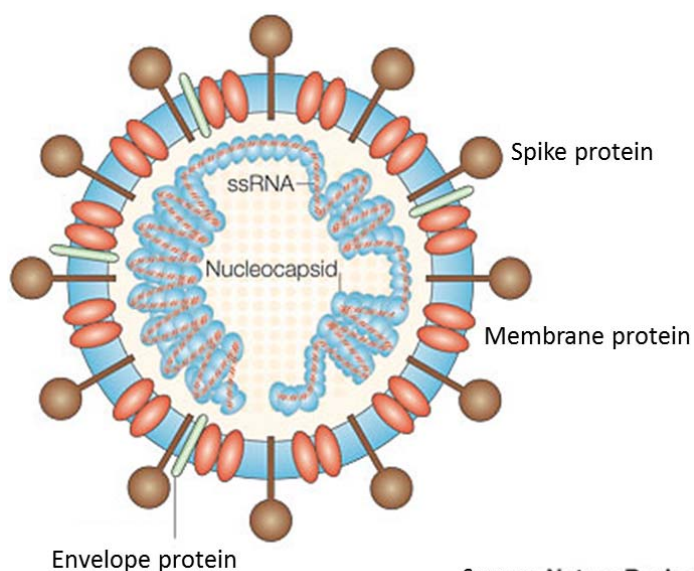
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[Total: 14]

QUESTION 5

In May 2014, the Middle East respiratory syndrome coronavirus (MERS-CoV), which was first reported in Saudi Arabia in 2012, infected two Americans who travelled to Saudi Arabia.

Coronaviruses are enveloped RNA viruses that infect and cause lower respiratory tract disease in a broad array of animals and humans. Virus particles range from 70 to 120 nm in diameter and are surrounded by characteristic spike-shaped glycoproteins, as shown in Fig. 5.1. Coronaviruses contain the largest single-stranded, positive-strand RNA genomes currently known, which range from 25.5 to nearly 32 kb in length.



Source: **Nature Reviews | Immunology**

Fig. 5.1

(a) Describe **two** structural differences between the genome of the coronavirus and the influenza virus. [2]

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(b) Describe how the coronavirus enters its host cell. [3]

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- (c)** Describe the process which allows the coronavirus to infect a *broad array of animals and humans* overtime. [2]

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- (d)** Unlike the human immunodeficiency virus, the coronavirus genome is not integrated into its host DNA.

Suggest how the coronavirus produces more copies of its genome. [2]

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- (e)** The fatality rate of coronavirus infections is approximately 60%.

Briefly explain how the coronavirus can cause death in humans. [1]

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[Total: 10]

QUESTION 6

- (a) Fig. 6.1 is an electron micrograph of a process that bacterial cells undergo which results in the formation of two daughter cells.

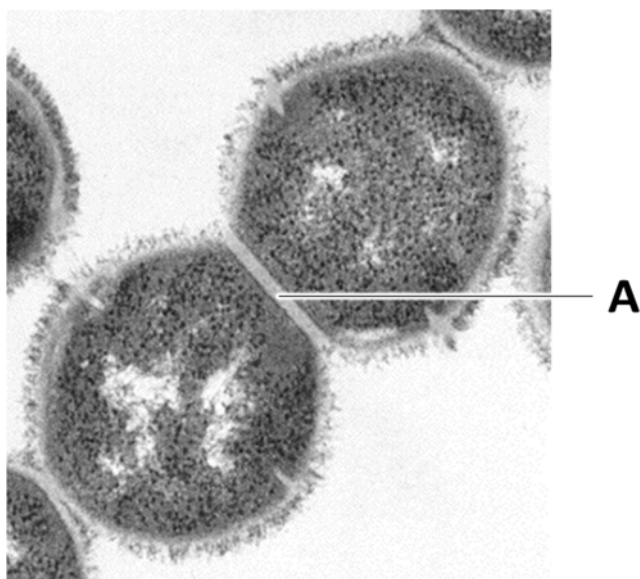


Fig. 6.1

- (i) Name the process above and state the main component making up structure **A**. [1]

Process

Component making up Structure A

- (ii) “The process above will always produce two genetically identical daughter cells”.

Comment on the validity of this statement. [1]

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- (b)** The *xyl* operon is a catabolic operon involved in the breakdown of the sugar xylose. Fig. 6.2 shows how a *xyl-lac* fusion operon is constructed, which consist of 2 structural genes from *lac* operon, regulatory sequences and the regulatory gene of the xylose operon. The arrows indicate the direction of transcription.

To test its effects, the fusion operon was constructed and packaged into bacteriophages. The fusion operon was then inserted into the chromosomes of these bacterial cells upon infection.

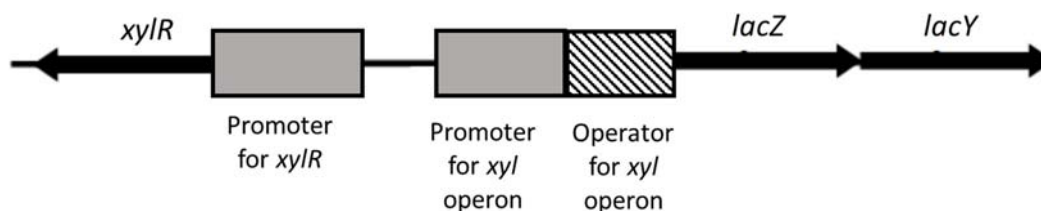


Fig. 6.2

- (i)** State the process of this gene transfer. [1]

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- (ii)** Suggest and explain one advantage of the process stated in **b(i)** over transformation in bacteria. [2]

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- (iii)** Explain the condition required for *lacZ* gene to be expressed in bacteria cells in which the *xyl-lac* fusion operon has been introduced. [3]

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- (iv)** Suggest why the direction of transcription of the regulatory and structural genes may differ. [2]

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- (c) Colibacillosis is a fatal condition caused by *E. coli* in poultry. In a study to examine the effectiveness of bacteriophages in treating colibacillosis, broiler chickens were first subjected to an aerosol spray containing bacteriophages on day 0. They were then separated into five treatment groups. Each treatment group was subsequently injected with *E. coli* on days 0, 1, 2, 3 and 4 respectively. The mortality rate for each treatment group was determined after 21 days. The result of the study is represented by Fig. 6.3 below.

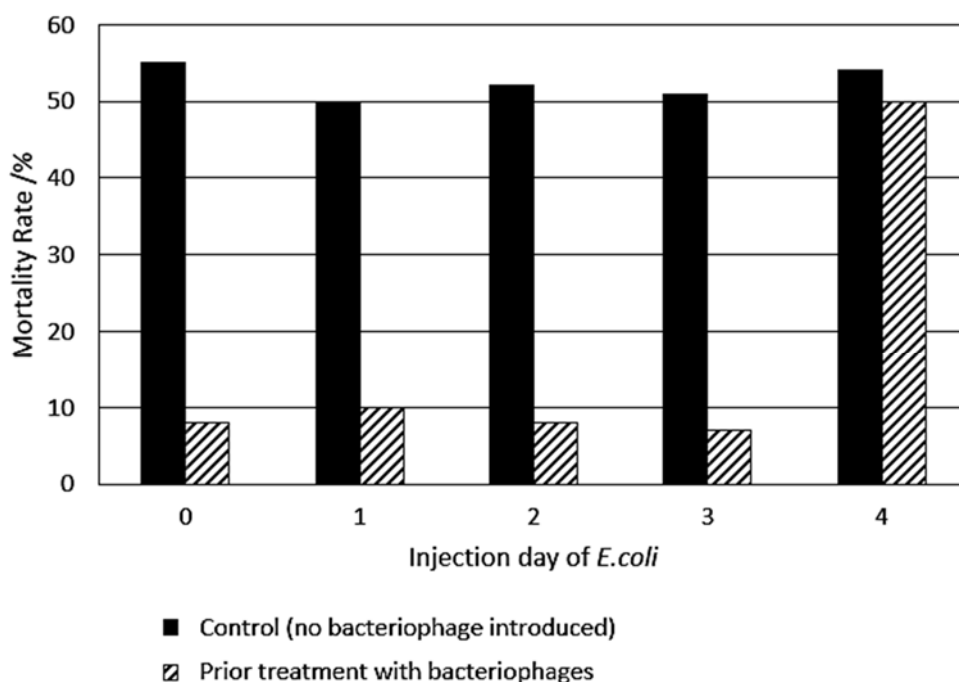


Fig. 6.3

With reference to Fig. 6.3 above,

- (i) Compare the trends observed in the control group and the groups that have been treated with bacteriophages, and comment on the effectiveness of such treatment. [3]

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- (ii) Suggest why the use of bacteriophages is a better alternative to antibiotic therapy for the chickens. [1]

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[Total: 14]

QUESTION 7

- (a) To analyse specific DNA sequences, various molecular processes are carried out. Fig. 7.1 below shows one possible way in which the gene for colour vision can be obtained for analysis.

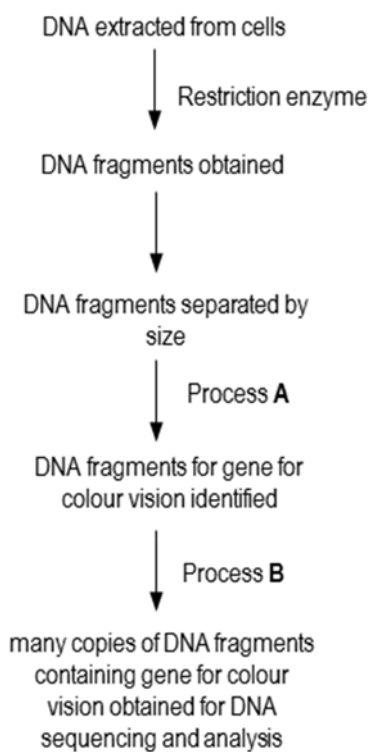


Fig. 7.1

- (i) Describe what is added to process **A** to identify the DNA fragment containing the gene for colour vision. [1]

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- (ii) Explain how process **B** ensures that many copies of the target sequence is produced. [1]

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(b) The inheritance of colour vision and ABO blood group was analysed in an extended family.

The gene for colour vision is sex-linked.

The gene for the ABO blood group system is on chromosome 9. There are three alleles controlling blood group. These three alleles give four possible phenotypes.

Fig. 7.2 shows the inheritance of these two genes in the extended family. Colour blindness is a rare condition, and can be assumed that the disease allele is not present in phenotypically normal individuals from other families.

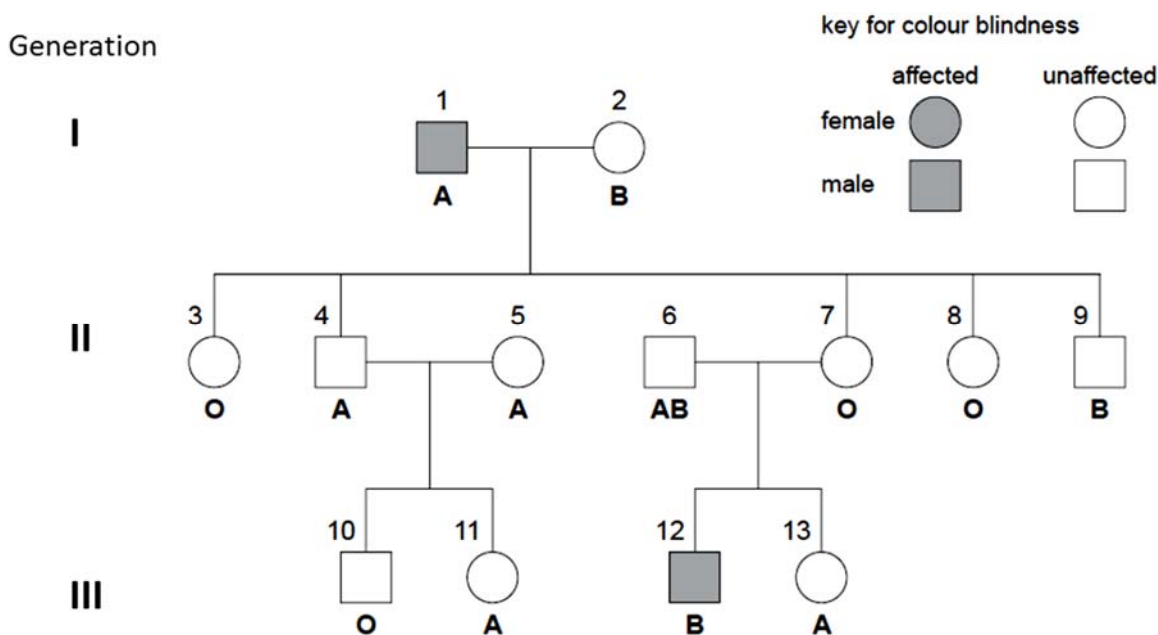


Fig. 7.2

(i) State a possible genotype for each of the following people in the family shown in Fig. 7.2. [2]

Individual **I-2**

Individual **II-9**

(ii) With reference to Fig. 7.2, explain why one grandson (**III-12**) of individual **I-1** has inherited colour blindness but the other (**III-10**) has not. [3]

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- (c) Nail-patella syndrome is a rare autosomal dominant trait that affects fingernails, toenails, elbows and kneecaps. The locus of the gene for nail-patella syndrome, **N** / **n**, is 10 map units from the ABO locus on chromosome 9.

A man with nail-patella syndrome and blood group AB has a family of five children with his wife who does have the syndrome and is blood group O.

Three children do not have the nail-patella syndrome and are blood group A.

Two children have nail-patella syndrome and are blood group B.

- (i) Use a genetic diagram to illustrate the above cross between the man and his wife. [3]

- (ii) Suggest why there is only a 5% probability of these parents having a child with both blood group A and nail-patella syndrome. [2]

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[Total: 12]

QUESTION 8

Fig. 8.1 shows how a rise in blood glucose concentration stimulates the beta cells in the pancreas to secrete insulin, a protein hormone.

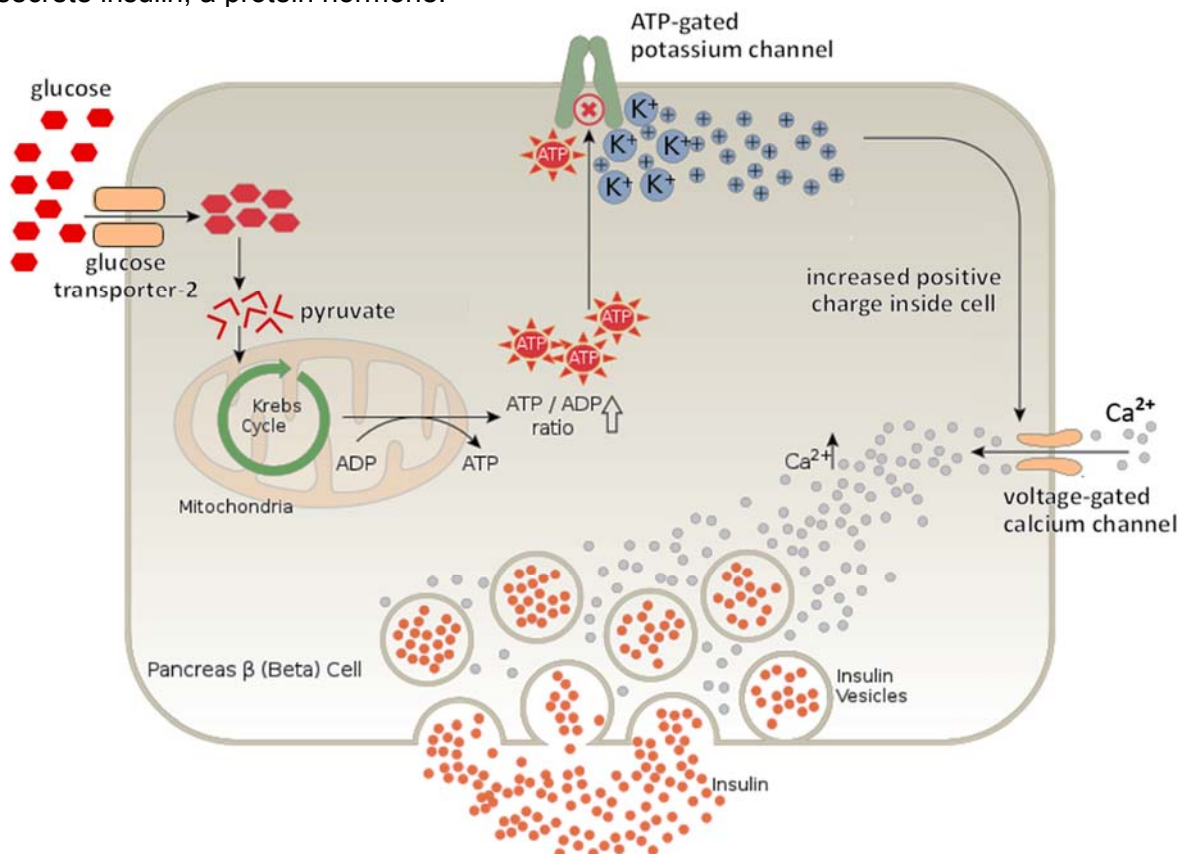


Fig. 8.1

- (a) Explain the significance of an existing pool of insulin-rich vesicles in the β-cell. [2]

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- (b) Outline the events leading to the release of insulin. [4]

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Insulin released by β -cells reaches their target cells, such as liver and muscle cells. One of the responses of insulin is glycogen synthesis, as shown in Fig. 8.2.

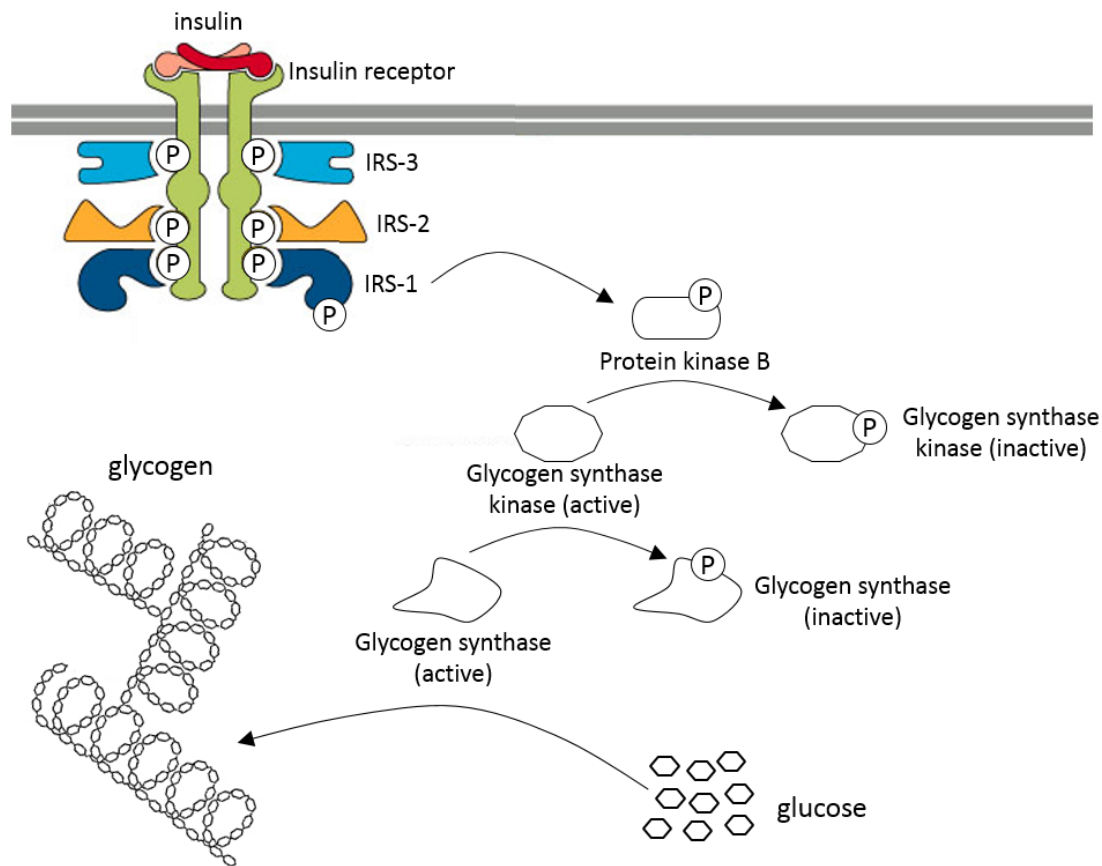


Fig. 8.2

(c) Describe how protein kinase B triggers glycogen synthesis. [3]

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[Total: 9]

QUESTION 9

A recent study of populations of the house mouse, *Mus musculus*, on the island of Madeira resulted in the following observations:

- There are six distinct populations.
- The mice are associated with human migration and settlements.
- The populations are located in different valleys separated by steep mountains.
- Each population has a different diploid number of chromosomes

As a result of these observations, it has been suggested that speciation is taking place.

Fig. 9.1 is a schematic representation of Madeira showing the distribution of the six populations.

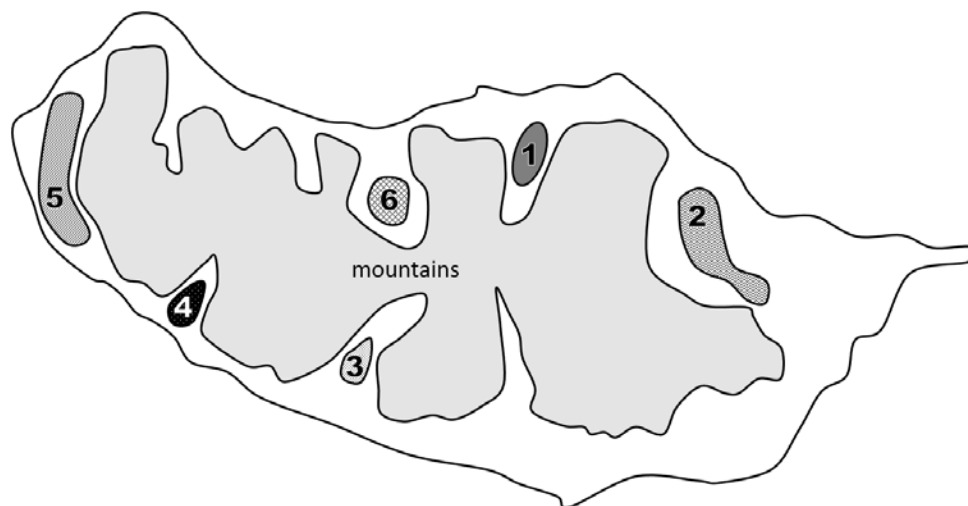


Fig. 9.1

- (a) Using the information in Fig. 9.1, state the likely isolating mechanism and the type of speciation taking place. [1]

isolating mechanism

type of speciation

(b) *'It has been suggested that speciation is taking place.'*

Explain how this process is occurring in the house mouse populations of Madeira. [5]

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(c) Explain the likely outcome of individuals from two separate populations being mated in captivity. [2]

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(d) House mouse is classified as class Mammalia, phylum Chordata, kingdom Animalia.

State one feature of the **cells** of the kingdom Animalia that distinguish them from the cells of other multicellular eukaryotes. [1]

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- (e) The evolutionary relationship between organisms is based on the hypothesis that the rate of mutation of DNA stays constant. The rate of mutation can be estimated by comparing the differences in amino acid sequences between species whose time of speciation is independently determined from the dating of fossils.

Explain why amino acid sequences of proteins could reveal useful evolutionary data for taxonomists. [2]

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[Total: 11]

• END OF PAPER •